YEAR 8 MATHS · CHAPTER 10 · MIND MAP

Transformations and congruence

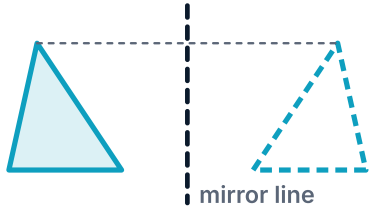
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1 Reflection

A mirror image: each point is the same distance from the mirror line.

$(2, 3)$ in the x-axis: $(2, -3)$

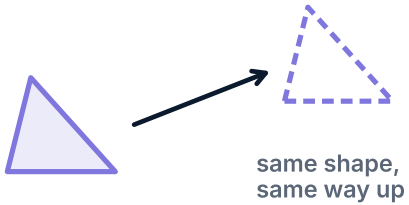
$(2, 3)$ in the y-axis: $(-2, 3)$



2 Translation

A slide, with no turning or flipping.

Right 3, down 2: $(1, 4) \to (4, 2)$



3 Rotation

A turn, given by a centre, an angle and a direction.

180° about the origin: $(2, 3) \to (-2, -3)$

90° clockwise about the origin: $(2, 3) \to (3, -2)$

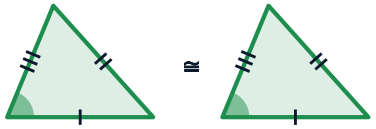


4 Congruent figures

Exactly the same shape and size ($\cong$). Reflections, translations and rotations keep figures congruent.

Matching sides and matching angles are equal

Write the vertices in matching order



5 Congruent triangles

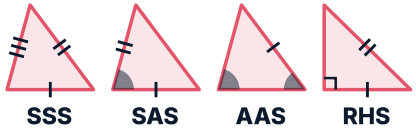
Tests: SSS, SAS, AAS, RHS.

SSS: three sides

SAS: two sides and the angle between them

AAS: two angles and a side

RHS: right angle, hypotenuse, side



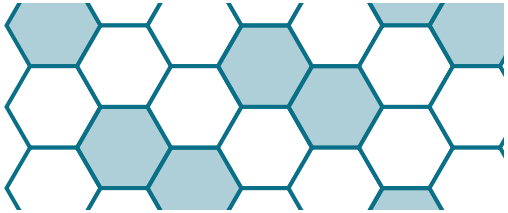
Watch out: AAA isn't a test: same angles can be different sizes.

6 Tessellations

Shapes that cover a surface with no gaps or overlaps. The angles at each point add to 360°.

Regular hexagons: $3 \times 120° = 360°$ ✓

Regular pentagons don't: 108° doesn't divide evenly into 360°

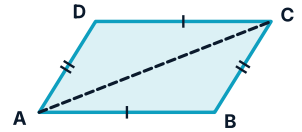


7 Congruence and quadrilaterals

A diagonal can split a quadrilateral into congruent triangles, which proves its properties.

A parallelogram's diagonal makes two congruent triangles, so opposite sides are equal

A rhombus's diagonals cross at right angles

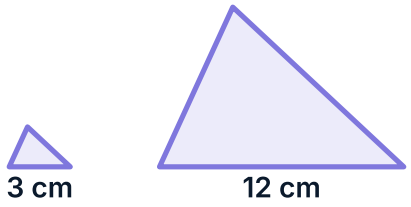


8 Similar figures

Same shape, different size. Scale factor $= \frac{\text{image length}}{\text{original length}}$.

3 cm becomes 12 cm: scale factor 4

The angles stay the same



9 Similar triangles

Same angles, sides in the same ratio. Use the scale factor to find missing sides.

Matching sides 3 and 12: scale factor 4, so side 5 matches 20

A 1 m stick has a 2 m shadow; a tree's shadow is 10 m: the tree is 5 m tall

